CAMBRIDGE INSTITUTE OF TECHNOLOGY
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MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY

Chips to Startup Programme

Cambrian Research Park

ChipIN

Comprehensive Electrical Characterization of NMOS Transistors in SCL 180 nm CMOS: I–V Behavior and Device Parameter Extraction

NMOS I–V Analysis and Characterization

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VI characteristics of transistor.

- For a MOSFET, the V–I characteristics describe how the drain current I_D changes with the applied gate-to-source voltage V_{GS} and drain-to-source voltage V_{DS}.
- For an NMOS, the two most important characteristics are:
- Transfer characteristic: I_D vs. V_{GS}
- Output characteristic: I_D vs. VDS

1. Transfer Characteristic — I_D vs VGS

While keeping V_{DS} sufficiently large/constant so the transistor operates in saturation.

The main purpose is to determine:

- Threshold voltage V_{TH}
- Drain current I_D
- Transconductance gm

2. Output Characteristic — I_D vs V_{DS}

For several fixed values of V_{GS}.

For example:

$$V_{GS} = 0.6V,\ 0.8V,\ 1.0V,\ 1.2V,\ 1.4V$$

We obtain a family of $I_D - V_{DS}$ curves.

The graph has three important regions.

Region	Condition	MOSFET State
Cutoff	$V_{GS} < V_{TH}$	OFF
Linear / Triode	$V_{DS} < V_{GS} - V_{TH}$	Voltage-controlled resistor
Saturation	$V_{DS} \geq V_{GS} - V_{TH}$	Current-control region

NMOS Behavior

When: $V_GS<V_TH$

The transistor is approximately OFF: $I_D\approx0$

When:$V_GS>V_TH$

An inversion channel forms and drain current starts increasing.

For the long-channel saturation model:

$$\boldsymbol{I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2}$$

where:

μ_n= electron mobility

C_ox= oxide capacitance per unit area

W= transistor width

L= transistor length

V_TH= threshold voltage

So the transfer curve approximately follows:

$$ID\propto(VGS-VTH)2$$

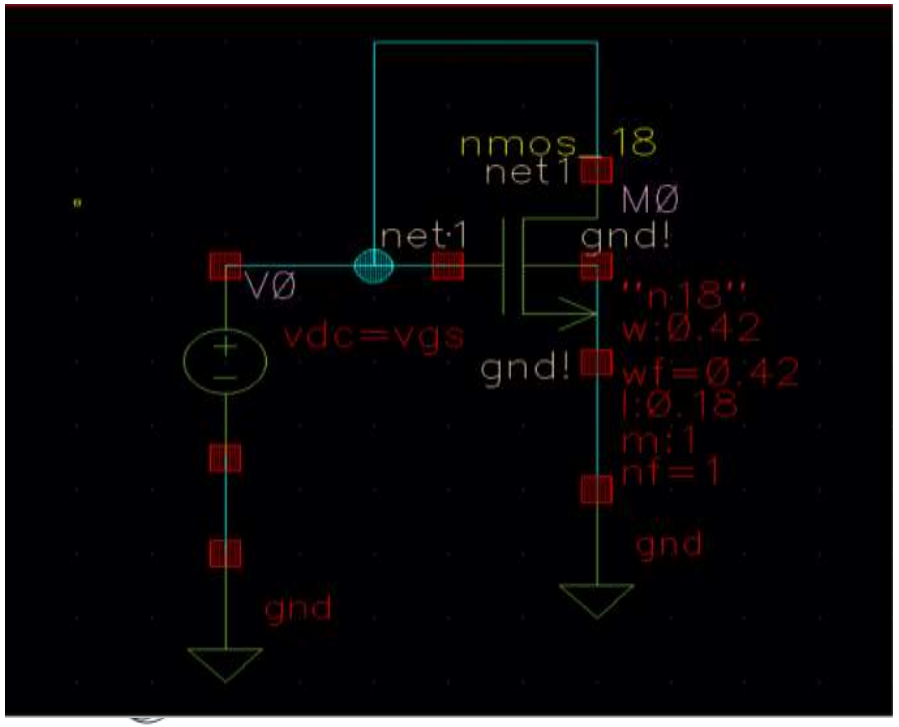
Without Bias Voltage (VSB = 0)

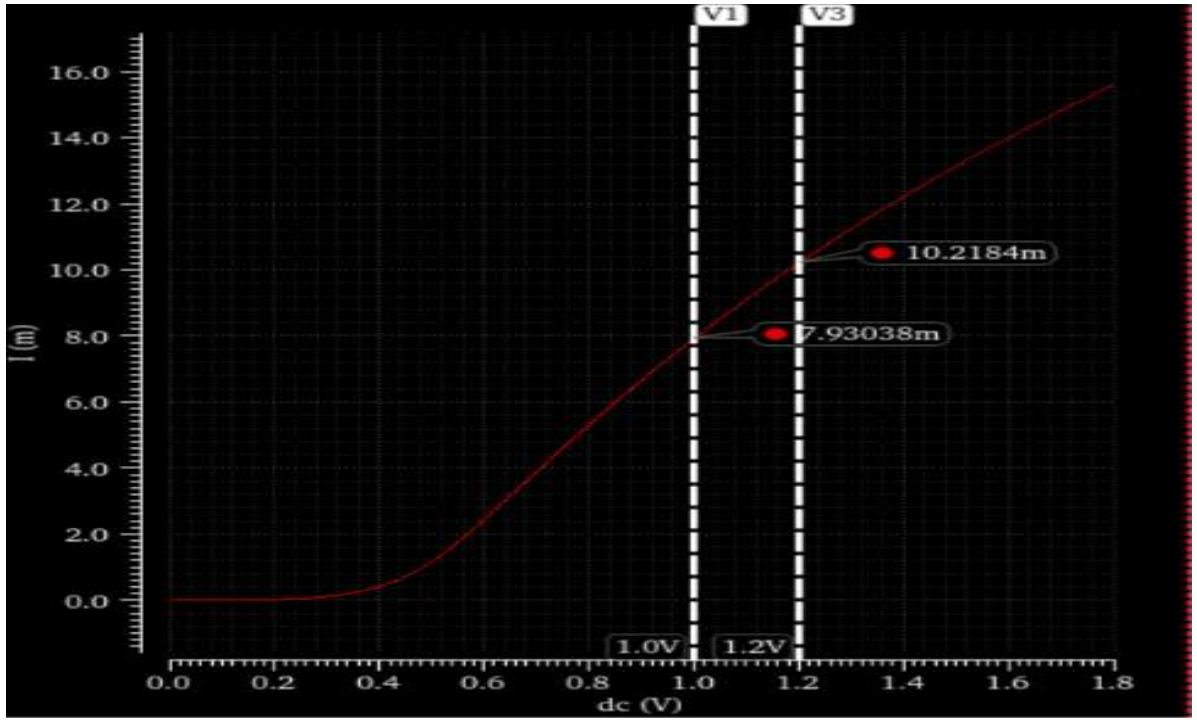
Connect gate and drain together So : $\boxed{V_D = V_G = V_{GS}}$

because source is at ground: $V_S = 0$

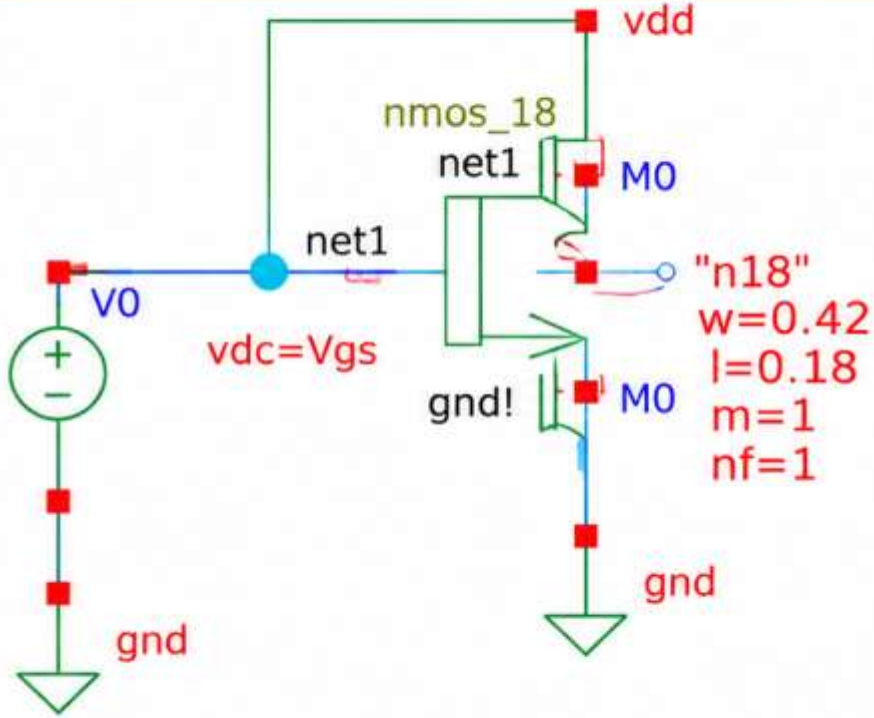
VDS=VGS

Parameters: Length- 180nm Width- 2um

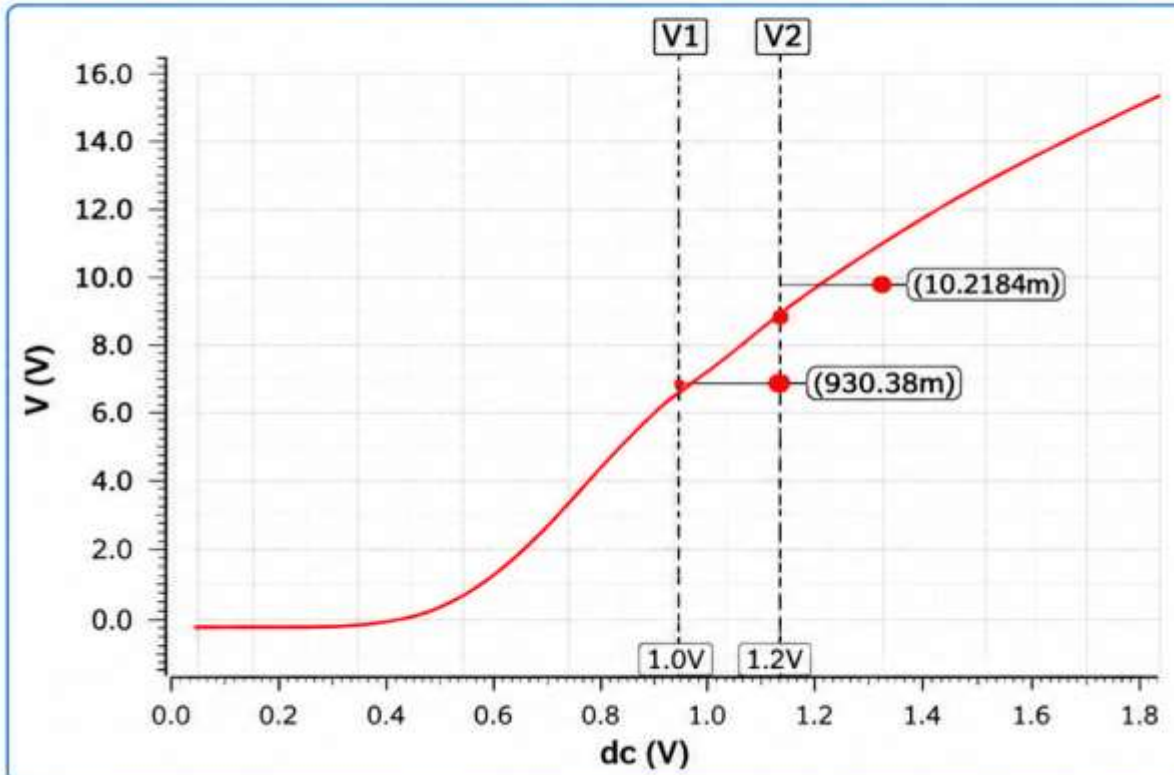




Circuit Schematic



DC Sweep Result (VTC)



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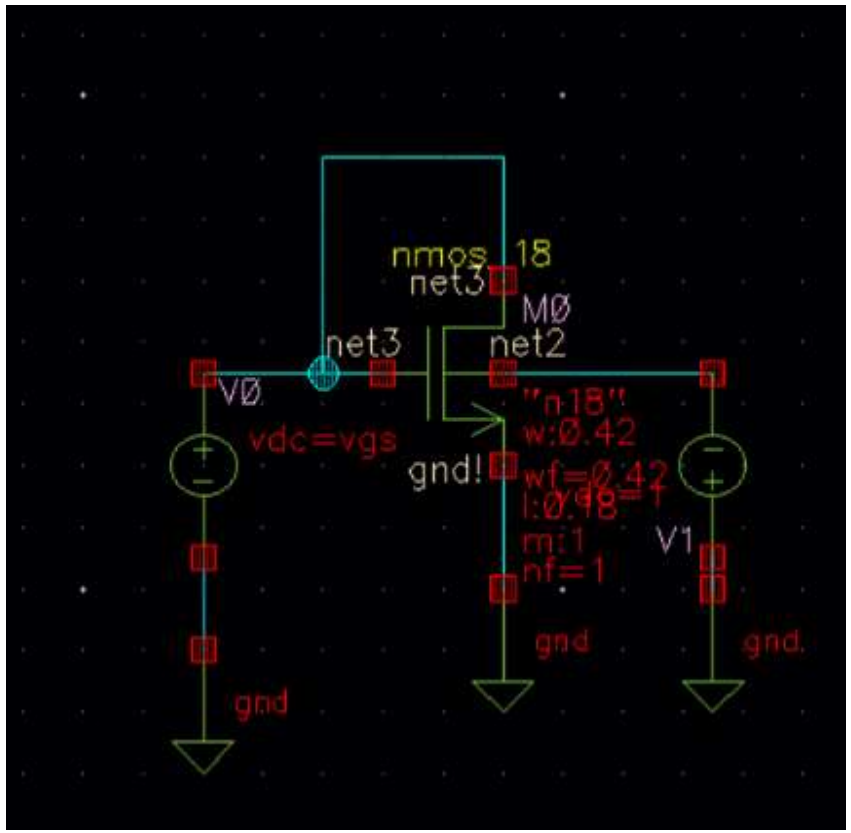
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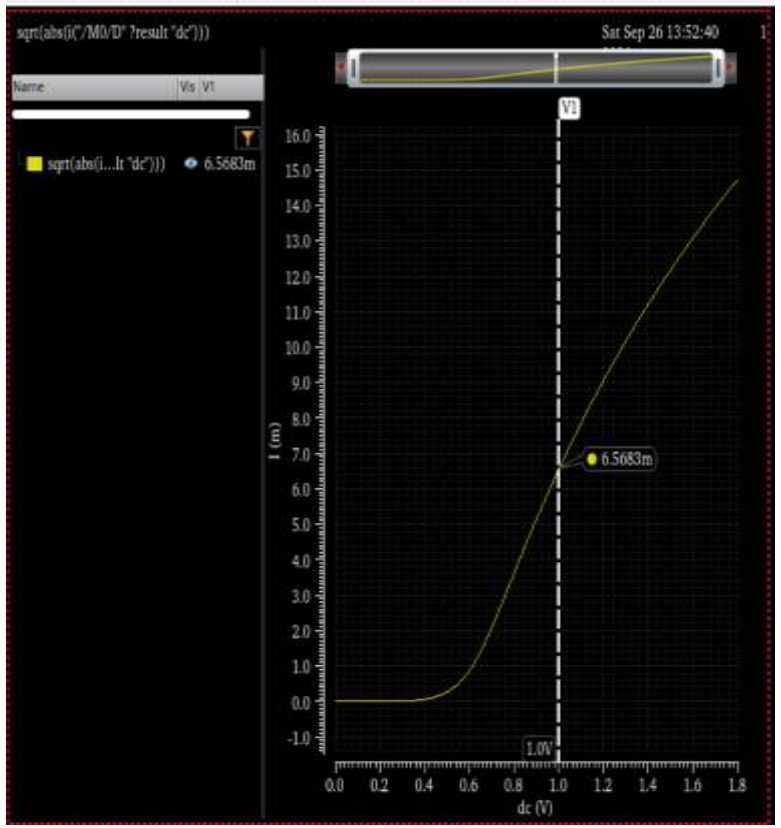
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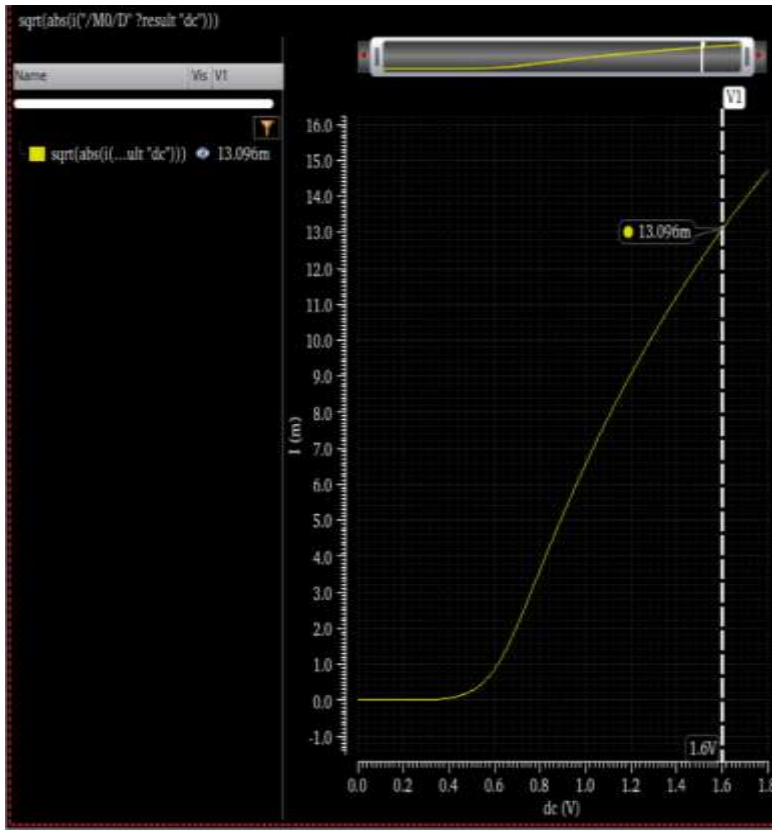
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C-DAC

With Bias Voltage (VSB = 1V)







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Connect gate and drain together

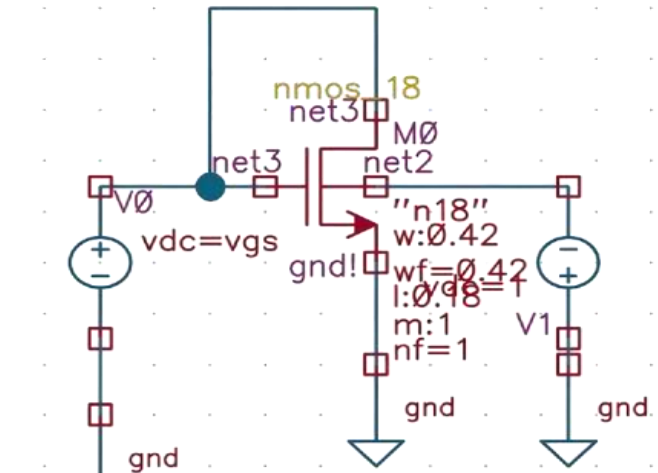
So : $\boxed{V_D = V_G = V_{GS}}$

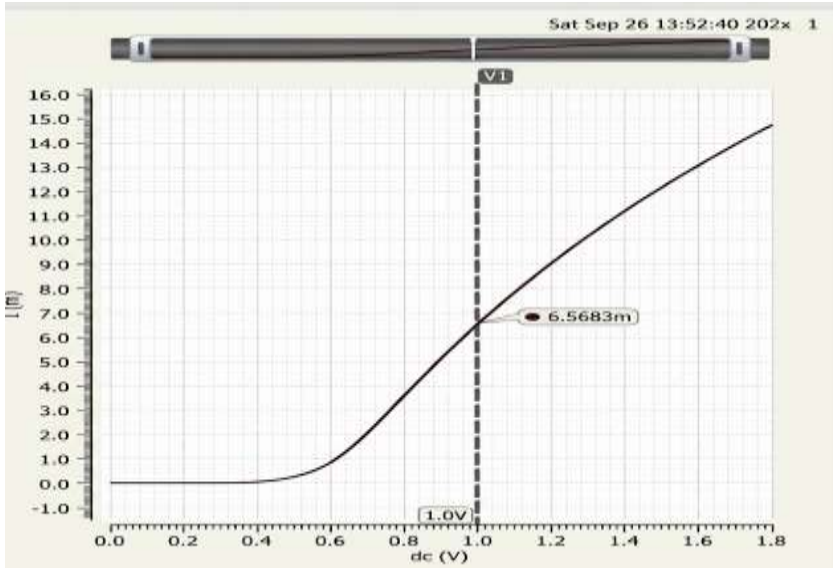
and because source is at ground:

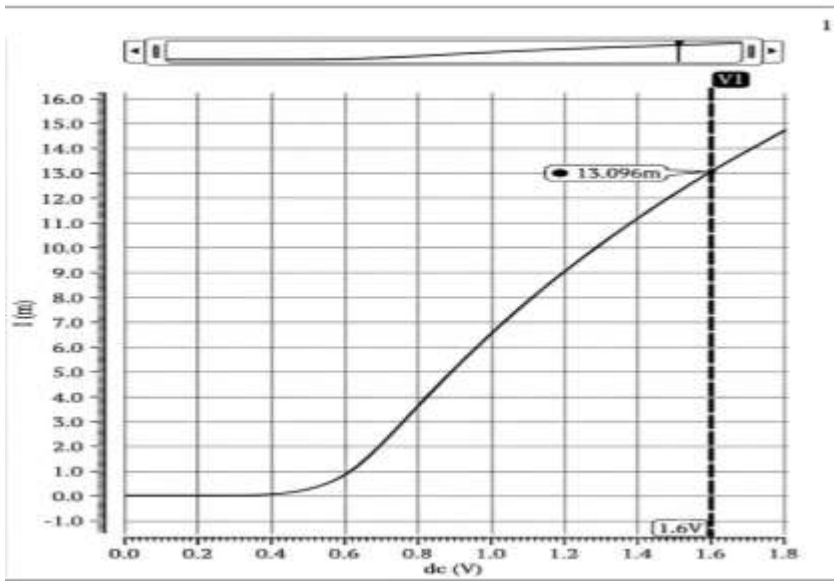
$$V_S = 0$$

therefore: VDS=VGS

Add Bias voltage at body VSB= -1V







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Body Effect Coefficient (γ)

From simulation

- $V_{TH0} = 0.428842\ V$ at $V_{SB} = 0$
- $V_{TH1} = 0.552229\ V$ at $V_{SB} = 1\ V$
- $2\phi_F = 0.6\ V$

Body-effect equation:

$$V_{TH} = V_{TH0} + \gamma\left[\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}\right]$$

Calculate γ:

$$\gamma = \frac{V_{TH1} - V_{TH0}}{\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}}$$

Substitute:

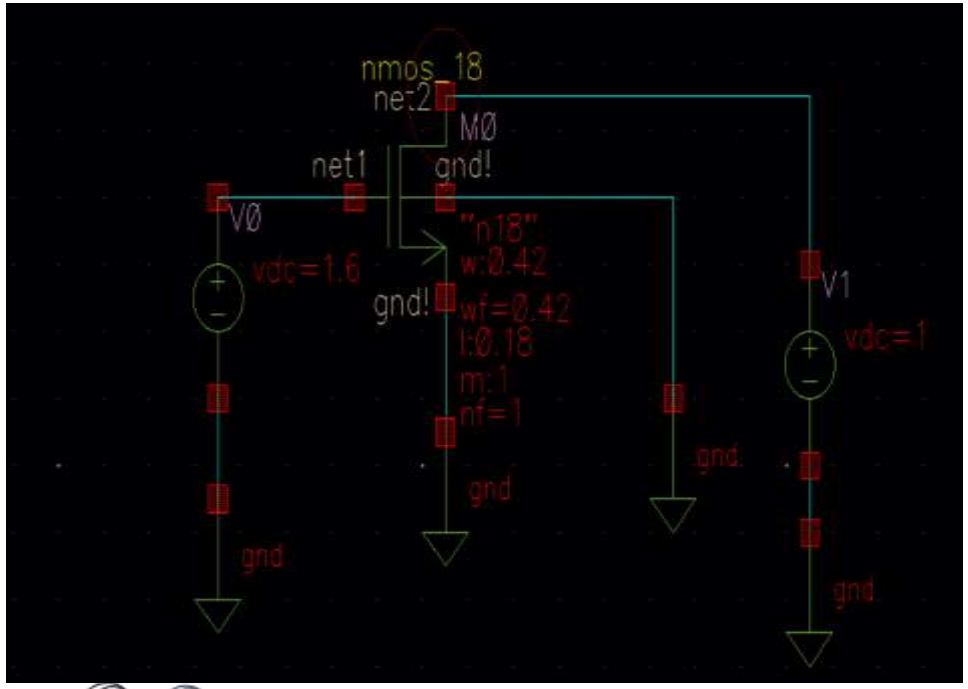
$$\gamma = \frac{0.552229 - 0.428842}{\sqrt{0.6 + 1} - \sqrt{0.6}}$$

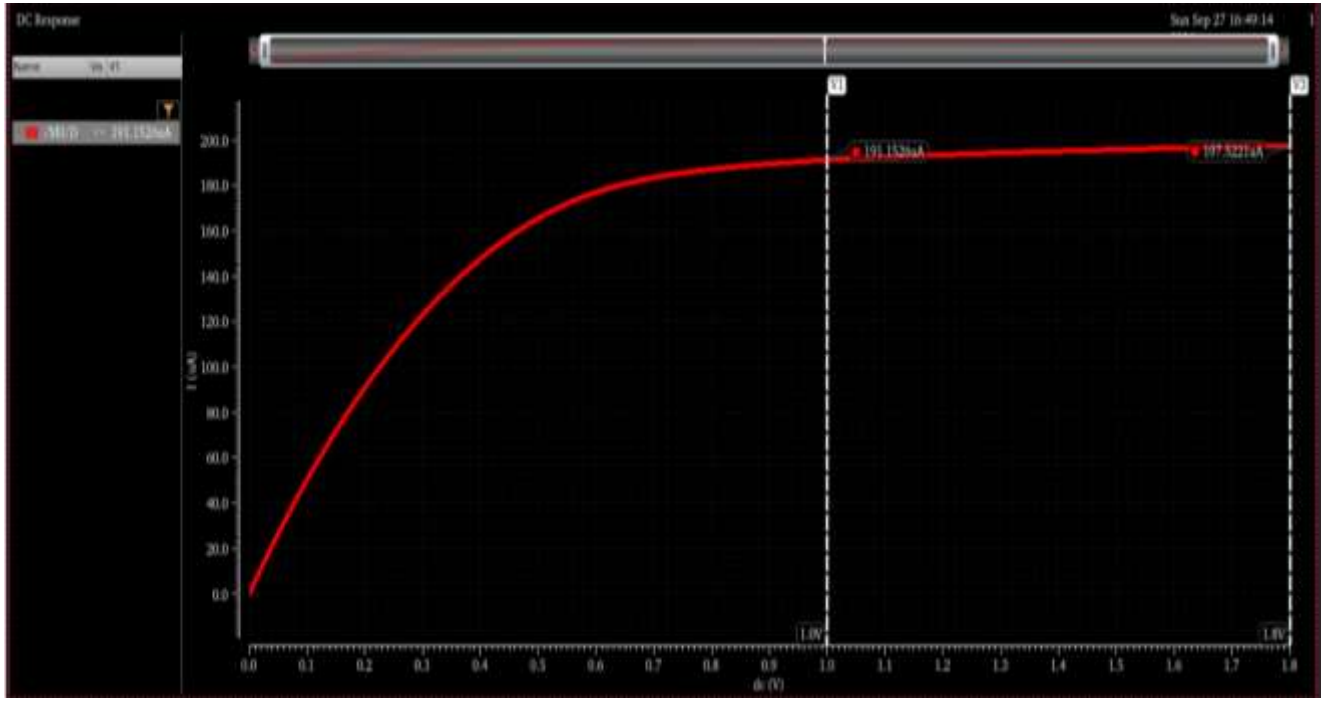
$$\gamma = \frac{0.123387}{1.2649 - 0.7746}$$

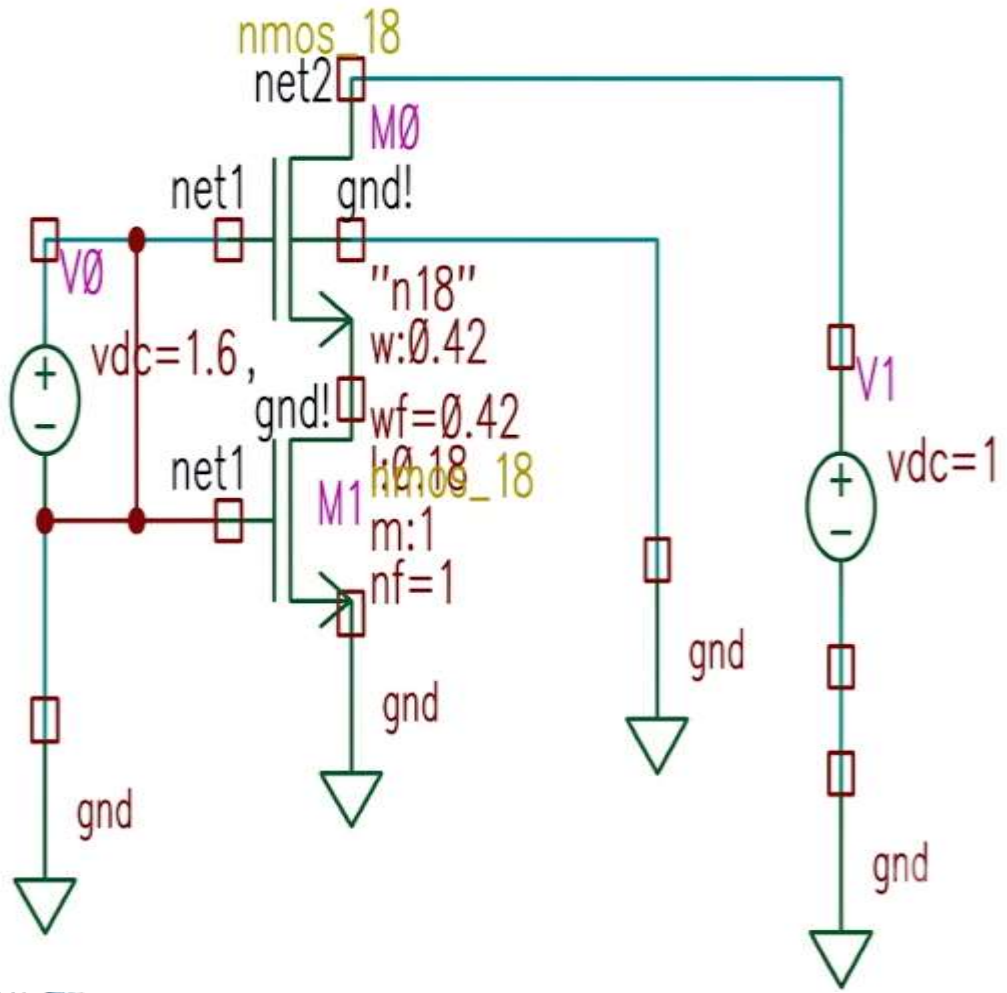
$$\boxed{\gamma \approx 0.2516\ V^{1/2}}$$

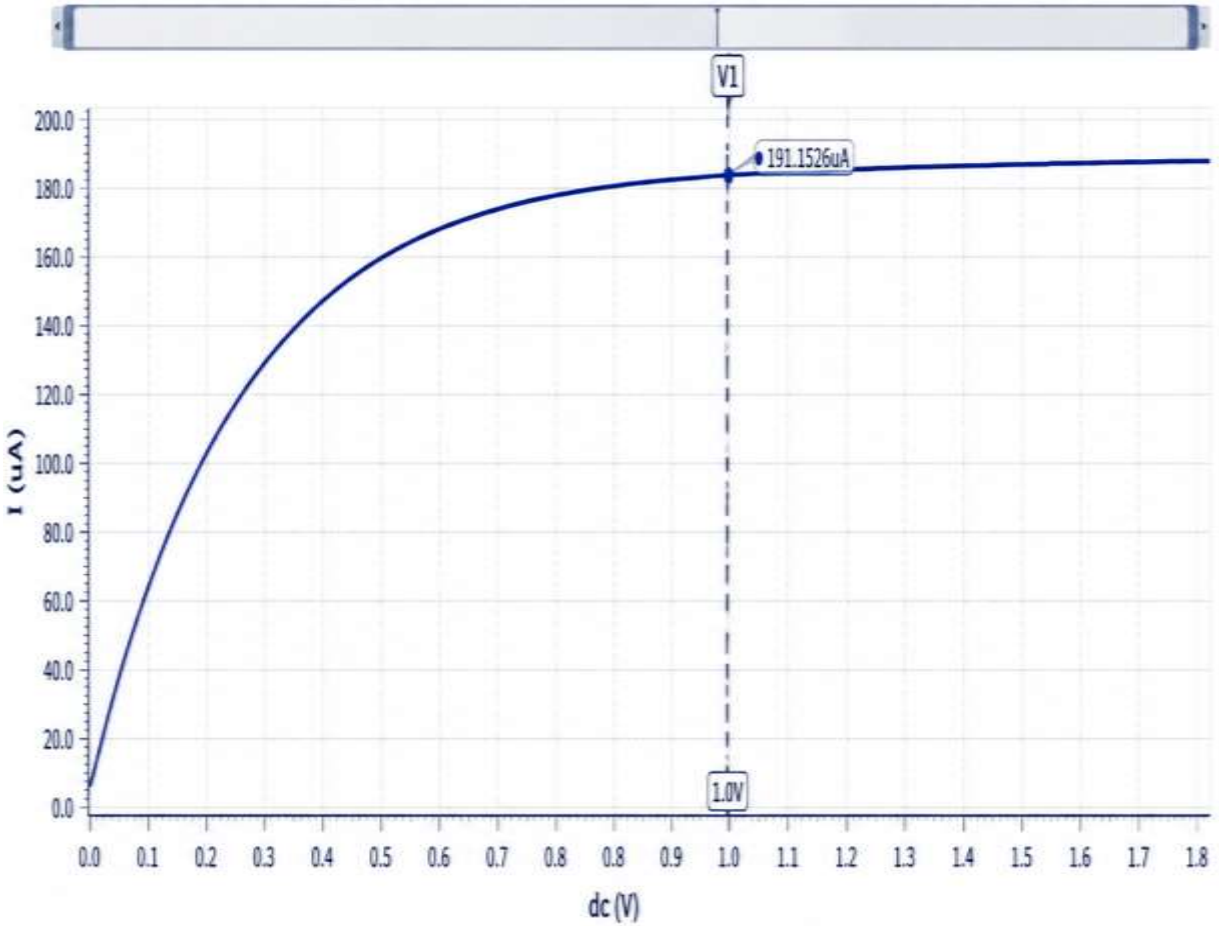
Channel Length Modulation

W.K.T VT0= 0. 6V
VGS=VT0+1
So VGS=0.6+1=1.6 V
Keep $V_{GS} = 1.6$ Vfixed.
Sweep V_{DS}.






nmos_18
net2
M0
net1
gnd!
V0
"n18"
w:0.42
vdc=1.6
gnd!
wf=0.42
net1
M1
nmos_18
m:1
nf=1
gnd
gnd
gnd
V1
vdc=1
gnd


V1
191.1526uA
1.0V
I (uA)
dc (V)
200.0
180.0
160.0
140.0
120.0
100.0
80.0
60.0
40.0
20.0
0.0
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.8 0.9 1.0 1.1 1.2 1.3 1.4 1.5 1.6 1.7 1.8

Channel Length Modulation – Extracted Results

At $V_{GS} = 0.8\,V$:

- $I_{D1} = 187.115\,\mu A$ at $V_{DS1} = 0.8\,V$
- $I_{D2} = 193.47\,\mu A$ at $V_{DS2} = 1.2\,V$

Using the two-point slope:

$$g_{ds} = \frac{\Delta I_D}{\Delta V_{DS}} = 15.89\,\mu S$$

$$r_o = \frac{1}{g_{ds}} = 62.95\,k\Omega$$

$$\boxed{\lambda = 0.0849\,V^{-1}}$$

VA=λ1=11.78V

Parameter	Extracted Value
V_{GS}	0.8 V
$I_{D1}@\ V_{DS} = 0.8V$	0.187115 mA
$I_{D2}@\ V_{DS} = 1.2V$	0.193470 mA
g_{ds}	15.8868 µS
r_o	62.9455 kΩ
λ	0.0849038 V^{-1}
$V_A = 1/\lambda$	11.778 V

- **To load the Design**

  - design("/home/m6/projects/Soumya/work_libs/m6/cds/simulation/nmos/spectre/schematic/netlist/netlist")

- **Specify Simulation Results Directory**

  - resultsDir("/home/m6/projects/Soumya/work_libs/m6/cds/simulation/nmos/spectre/schematic")

- **Select the SCL 180 nm NMOS Model**

  - path("/opt/sclpdkv3/scl180/pdk/cdns/sclpdk_v3//HOTCODE/amslibs/cds_default/cdslibs/ts18scl/models/default/spectre")

  - modelFile(("/opt/sclpdkv3/scl180/pdk/cdns/sclpdk_v3/HOTCODE/models/ts18scl/v2.0/hspice/ts18sl_scl.lib" "tt_18"))

➢ **To Perform V_{GS}DC Sweep**

Sweep VGS (V0) from 0 to 1.8V
analysis('dc
?saveOppoint t
?dev "/V0"
?param "dc"
?start "0"
?stop "1.8"
?step "0.01"
)

➢ **Set Initial V_{GS}Design Variable**

desVar("vgs" 0.6)

➢ **Define Analysis Order and to select NMOS Drain Current**

- envOption('analysisOrder list("dc"))
- save('i "/M0/D")

➢ **Set Simulation Temperature -** temp(27)

➢ **To run the Simulation and to read the DC Simulation Result**

- run()
- selectResult('dc)
  id0_vgs = getData("/M0/D")

➢ **Calculate$\sqrt{I_D}$**

- sqrtid0 = sqrt(abs(id0_vgs))

- **To plot the waveforms**

```
Open a single new window
win = newWindow()

; Subwindow 1: Body Effect Curves (sqrt(ID) vs VGS)
plot(sqrtid0 ?expr '("sqrt(ID) [VSB = 0V]"))
plot(sqrtid1 ?expr '("sqrt(ID) [VSB = 1V]"))

; Subwindow 2: Channel Length Modulation Curve (ID vs VDS)
addSubwindow()
plot(id_vds ?expr '("ID vs VDS [VGS = 0.8V]"))
```

- **Extract √ID values**

sqrt_id_0_low = value(sqrtid0 0.6)
sqrt_id_0_high = value(sqrtid0 0.8)

sqrt_id_1_low = value(sqrtid1 0.8)
sqrt_id_1_high = value(sqrtid1 1.0)

sqrtid0 → √ID curve for VSB = 0 V
sqrtid1 → √ID curve for VSB = 1 V

- **Calculate γ**

2 ϕ_F = 0.60
VSB = 1.0

gamma = (VTH1 - VTH0) /
(sqrt(2 ϕ_F + VSB) - sqrt(2 ϕ_F F))

$$\gamma = \frac{V_{T1} - V_{T0}}{\sqrt{2 \mid \phi_F \mid + V_{SB}} - \sqrt{2 \mid \phi_F \mid}}$$

- **Calculate slopes**

slope0 = (sqrt_id_0_high - sqrt_id_0_low) / 0.2
slope1 = (sqrt_id_1_high - sqrt_id_1_low) / 0.2
m= $\sqrt{I_{D2}}$- $\sqrt{I_{D1}}$ ÷ V_{GS2}- V_{GS2}

Complete Extraction Flow

OCEAN Script

$$\boxed{\text{Load NMOS} \rightarrow \text{Select SCL 180 nm Model} \rightarrow V_{GS}: 0 \rightarrow 1.8V}$$

$$\downarrow$$

$$\boxed{I_D(V_{GS}) \rightarrow \sqrt{|\ I_D\ |} \rightarrow \sqrt{I_D} \text{ vs } V_{GS}}$$

$$\downarrow$$

$$\boxed{V_{T0}}$$

Then, for the second simulation:

$$V_{SB} > 0$$

$$\downarrow$$

$$\boxed{V_{T1}}$$

and finally

$$\boxed{\gamma = \frac{V_{T1} - V_{T0}}{\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}}}$$

This gives the complete **body-effect coefficient extraction procedure**.

Calculation of channel length modulation

➢ **Define VDS points**

vds1 = 0.8

vds2 = 1.2

➢ **Extract drain currents**

id1 = value(id_vds vds1)

id2 = value(id_vds vds2)

➢ **Calculate output conductance**

gds = (id2 - id1) / (vds2 - vds1)

➢ **Extract λ**

lambda = gds / id1

Using:

$$g_{ds} \approx \lambda I_D$$

Therefore,

λ=IDgds

➢ **Calculate Early voltage**

VA = 1.0 / lambda

$$\boxed{V_A = \frac{1}{\lambda}}$$

Key Result

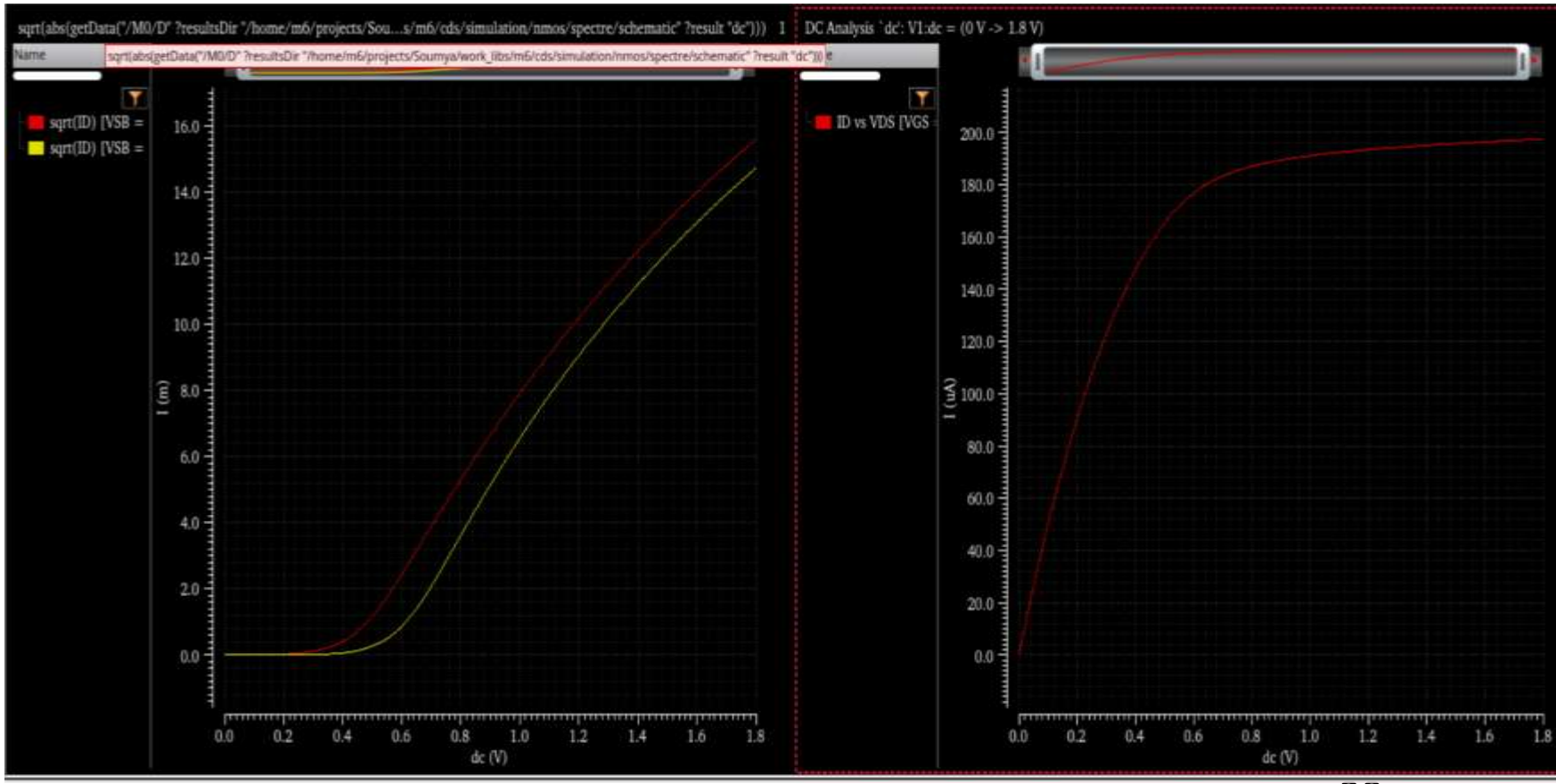
$$\boxed{I_D \uparrow \text{ with } V_{DS} \text{ in saturation}}$$

$$\boxed{\lambda = \frac{g_{ds}}{I_D}} \quad \boxed{V_A = \frac{1}{\lambda}}$$

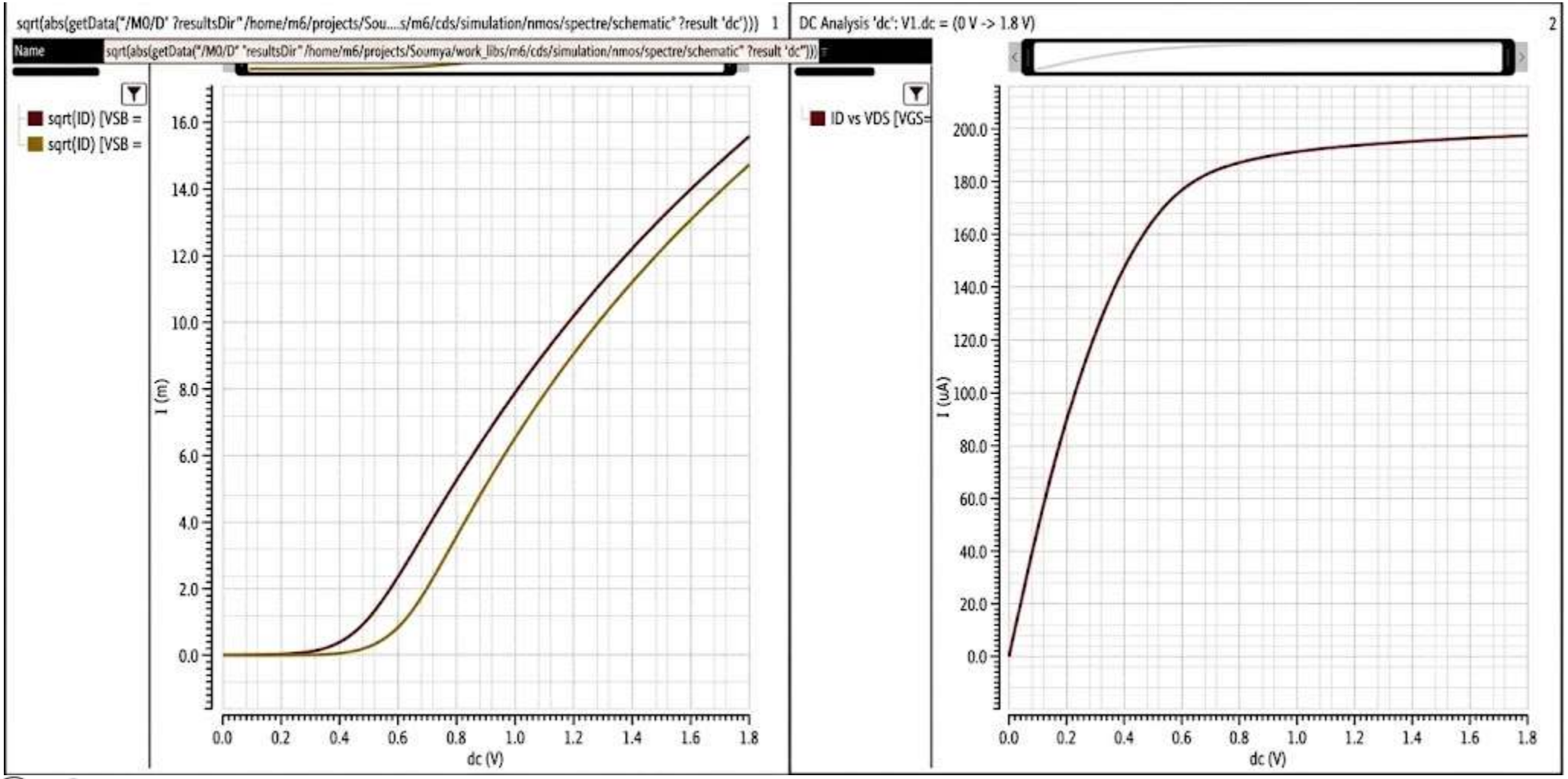
Outputs of body effect coffecient (γ) and channel length modulation(λ)

```
================================================
        PARASITIC PARAMETER RESULTS
================================================
1. BODY EFFECT PARAMETERS (GAMMA)
   ------------------------------------------------
   Slope (VSB = 0 V) : 0.0143513
   Slope (VSB = 1 V) : 0.0146718
   VTH0 (VSB = 0 V)  : 0.428842 V
   VTH1 (VSB = 1 V)  : 0.552229 V
   2PhiF             : 0.6 V
   GAMMA (gamma)     : 0.251648 V^(1/2)

2. CHANNEL LENGTH MODULATION PARAMETERS (LAMBDA)
   ------------------------------------------------
   VDS1 = 0.8 V  ==>  ID1 = 0.000187115 A
   VDS2 = 1.2 V  ==>  ID2 = 0.00019347 A
   Output Resistance (rout) : 62945.5 Ohms
   LAMBDA (lambda)          : 0.0849038 V^-1
   Early Voltage (VA)       : 11.778 V
================================================
t
```



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sqrt(abs(getData("/M0/D" ?resultsDir "/home/m6/projects/Sou....s/m6/cds/simulation/nmos/spectre/schematic" ?result 'dc'))) 1
Name
sqrt(abs(getData("/M0/D" "resultsDir" "/home/m6/projects/Soumya/work_libs/m6/cds/simulation/nmos/spectre/schematic" ?result "dc")))
sqrt(ID) [VSB =
sqrt(ID) [VSB =
I (m)
dc (V)
DC Analysis 'dc': V1.dc = (0 V -> 1.8 V) 2
ID vs VDS [VGS=
I (uA)
dc (V)

Comparison Table

Parameter	Experimental value	Theoretical value
Threshold voltage V_{TH0}	0.428842 V	vth0 = 0.377942 V
ro	62.9455	---
Body effect Coefficient (Gamma)	0.251648 V^(1/2)	k1=0.569482 k2=−0.0501868
Channel length modulation(lambda)	0.0849038 V^-1	pclm = 1.00521